ScaDS.AI DRESDEN LEIPZIG
CENTER FOR SCALABLE DATA ANALYTICS AND ARTIFICIAL INTELLIGENCE

come2data
COMPETENCE CENTER FOR INTERDISCIPLINARY DATA SCIENCE

NFDI4 BIOIMAGE
NATIONAL RESEARCH DATA MANAGEMENT INFRASTRUCTURE FOR MICROSCOPY AND BIOIMAGE ANALYSIS

Legal Framework

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Chapter 1:
Ethics, Benefits and Risks

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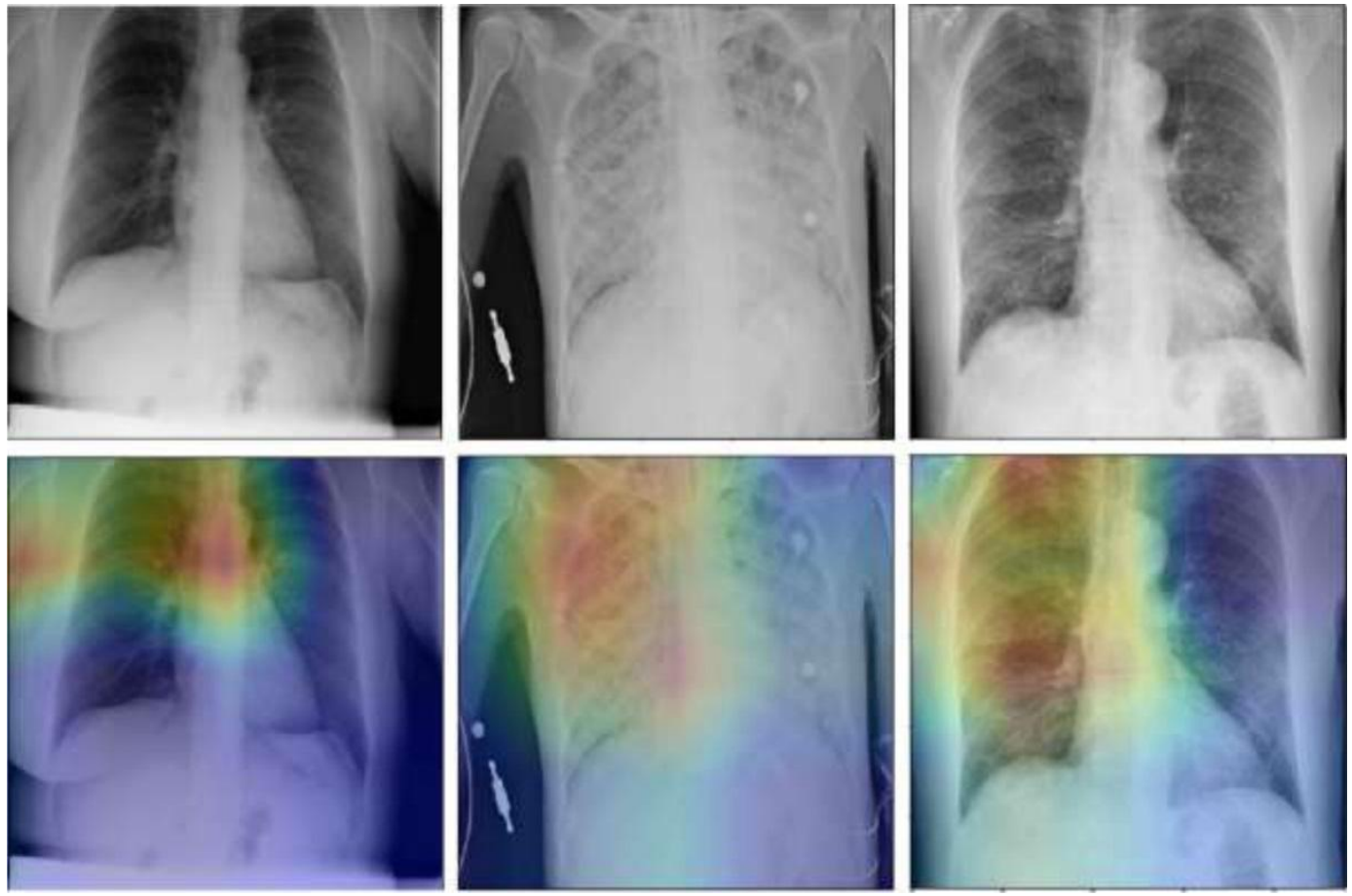

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Comparison of Accurate X-Ray Image Analysis for COVID-19

- **AI**-assisted analysis**: 97%**
- Conventional analysis by **radiologists: 53%**

Chapter 1: Ethics, Benefits and Risks

COMPAS („Correctional Offender Management Profiling for Alternative Sanctions“)

An algorithm that uses 137 characteristics to calculate the **probability of whether an offender will reoffend or not**.

Judges in the U.S. use COMPAS, among other things, to decide whether someone should be released on **probation** or remain in **custody**.

The algorithm is proprietary, so it is unknown exactly how it works. COMPAS is a black box.

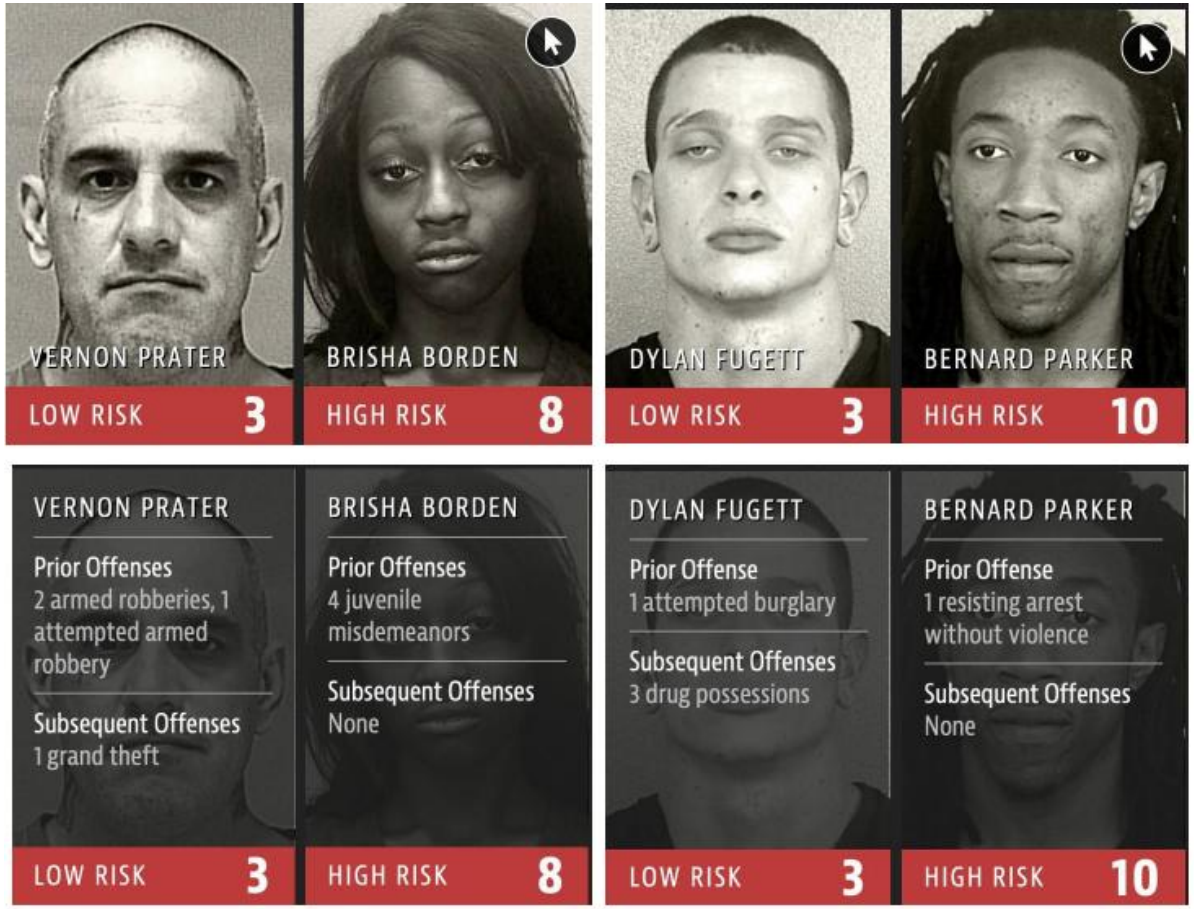

Image: Angwin, J., Larson, J., Mattu, S., & Kirchner, L. "Machine Bias: Risk Assessments in Criminal Sentencing." Veröffentlicht auf ProPublica, 23. Mai 2016. URL: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

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Chapter 1: Ethics, Benefits and Risks

Benefits of AI:

- Improvement of everyday processes in both personal and professional contexts
- Creation of new opportunities for freedom through innovative technologies (e.g., autonomous driving)

Risks of AI:

- **Individual** Harms
  - **Bias** and discrimination
  - Civil rights and privacy concerns
- **Societal** Harms
  - Group Discrimination
  - Mass Surveillance
  - Echo Chambers
- **Environmental** Harms
- **Organizational** Harms

Chapter 1: Ethics, Benefits and Risks

➡️ **Legal regulation** as a tool to **amplify the benefits** and **mitigate risks**

EU Approach

↓

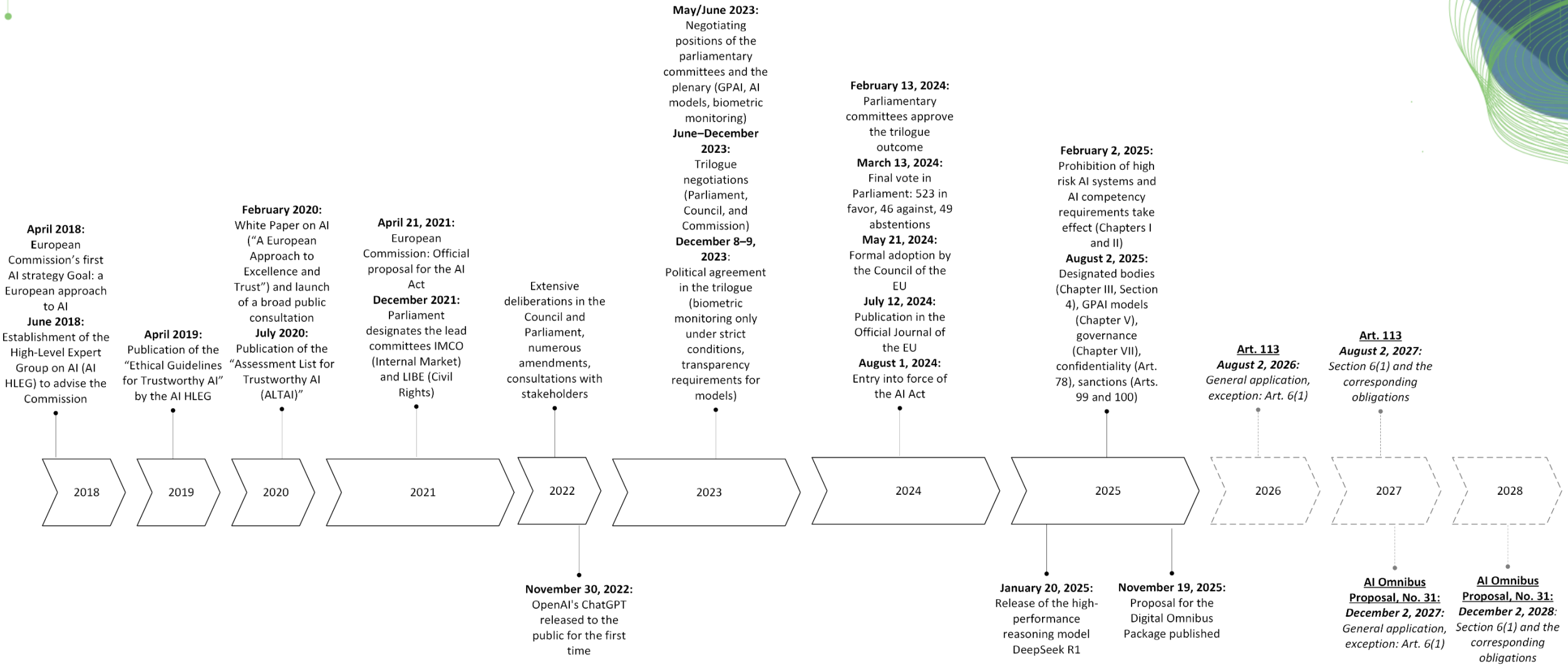
Value-based AI

“human centric and trustworthy”

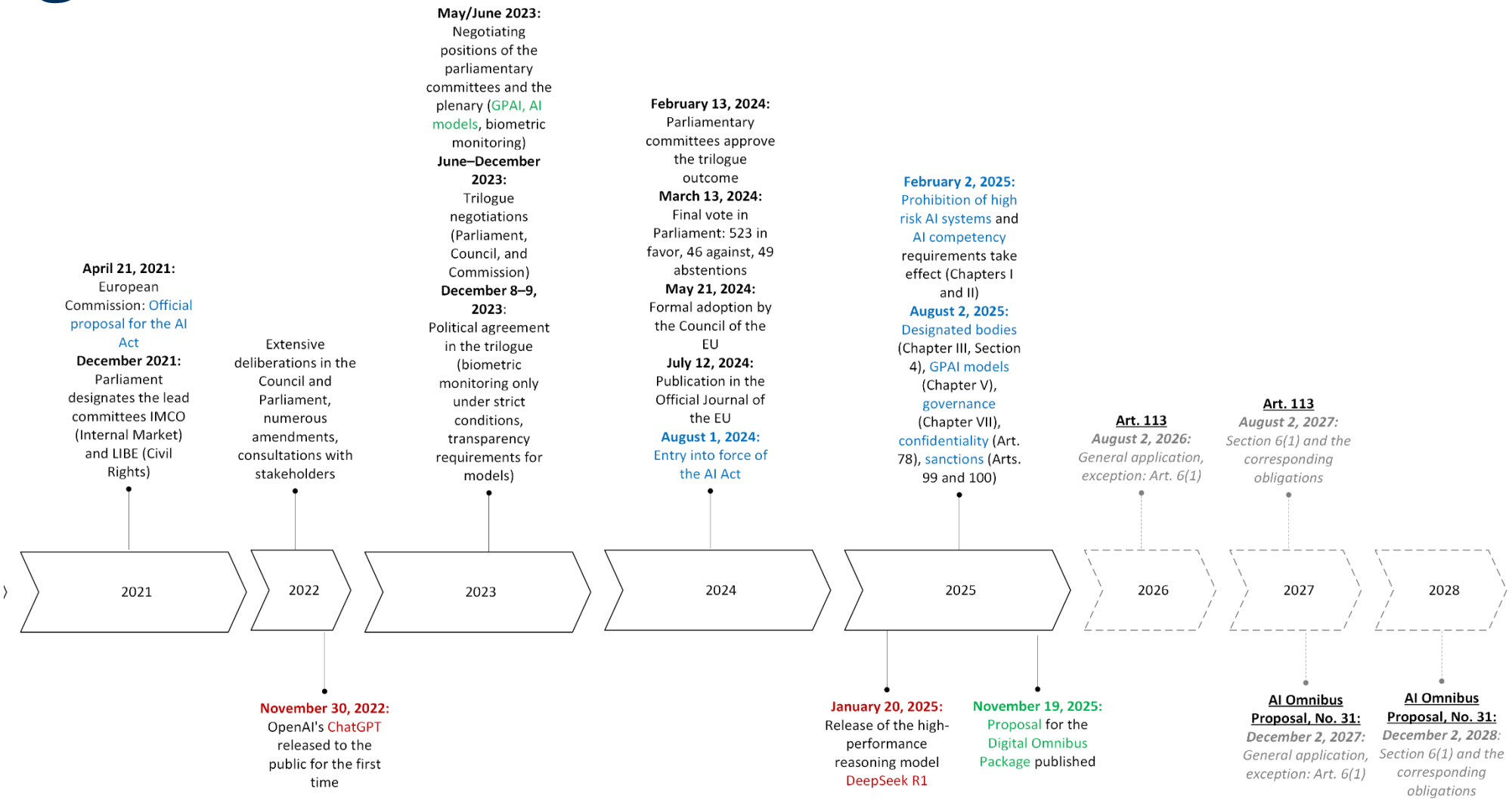
(Rec. 1)

Chapter 2:
Introduction to the AI Act

Regulation (EU) 2024/1689



Regulation (EU) 2024/1689



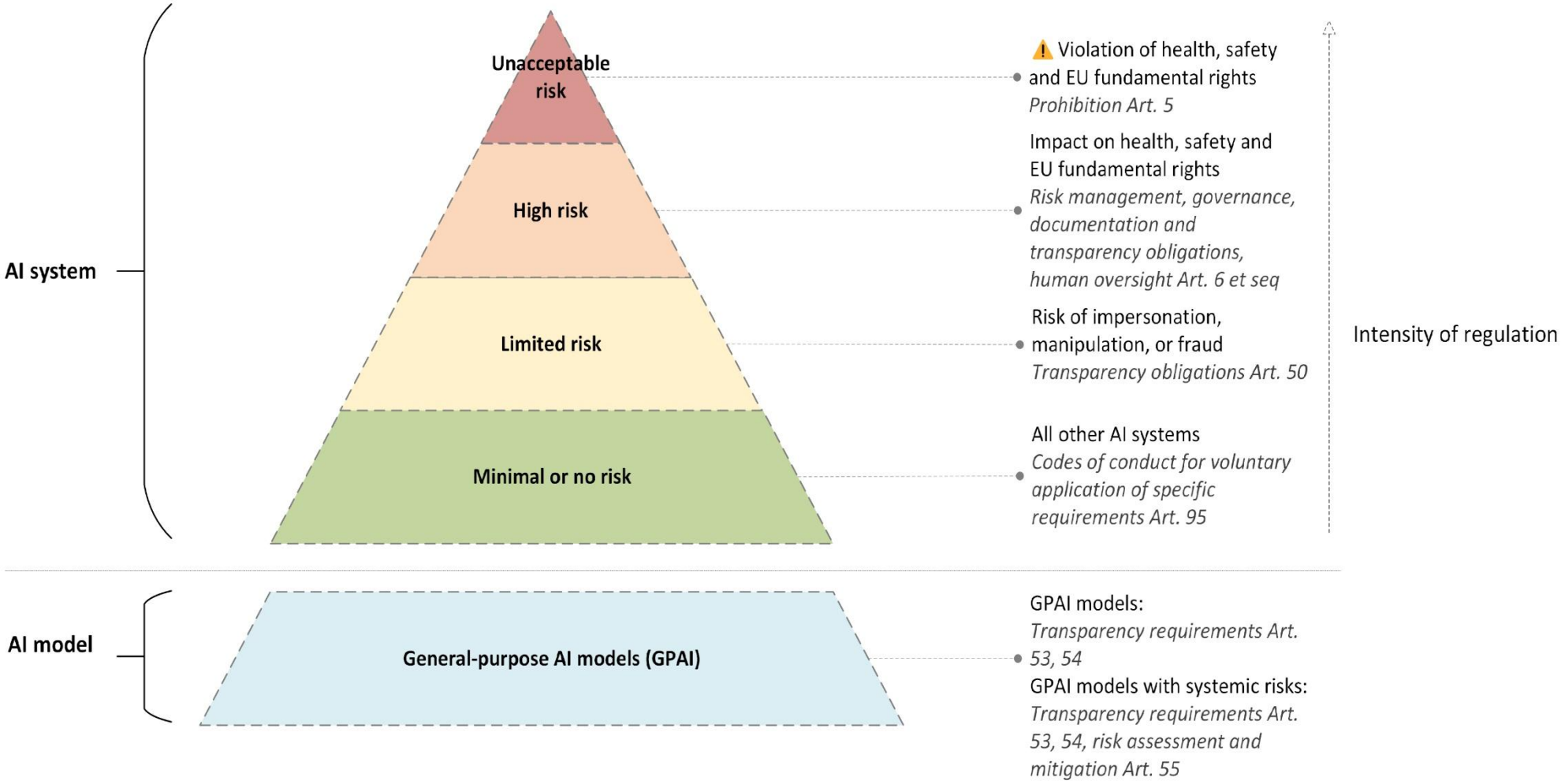
AI system (Art. 3 para. 1)

- a machine-based system
- operates with varying levels of autonomy
- adaptiveness after deployment
- infers, from the input it receives, how to generate outputs
- E.g. **ChatGPT**

AI model (Rec. 97)

- algorithm or computational framework ➡️ essential components of AI systems
- For AI models to become AI systems, additional components, such as a user interface, must be added.
- E.g. **GPT-4**

Risk based approach



Chapter 3:
Other Regulatory Framework

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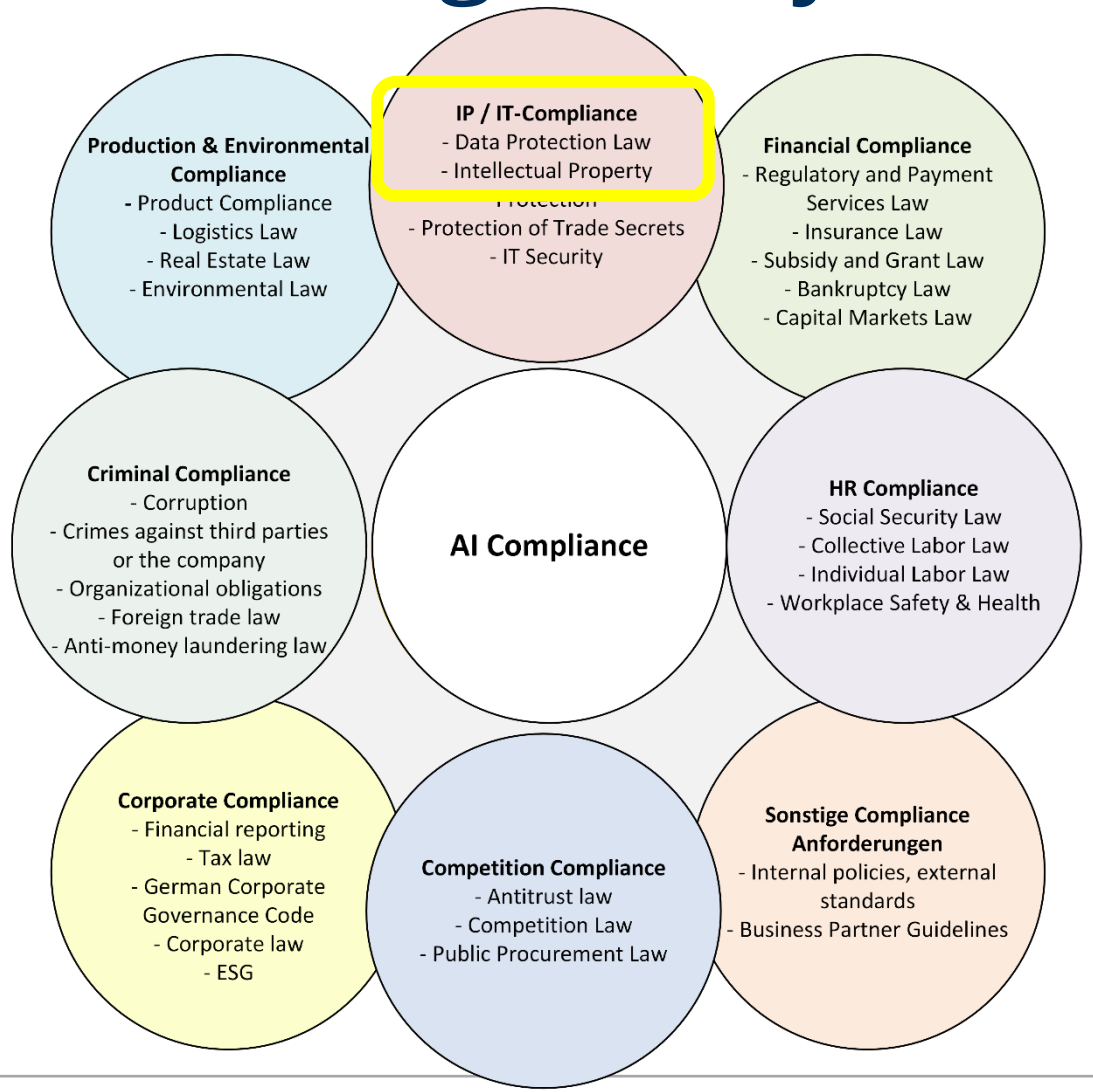
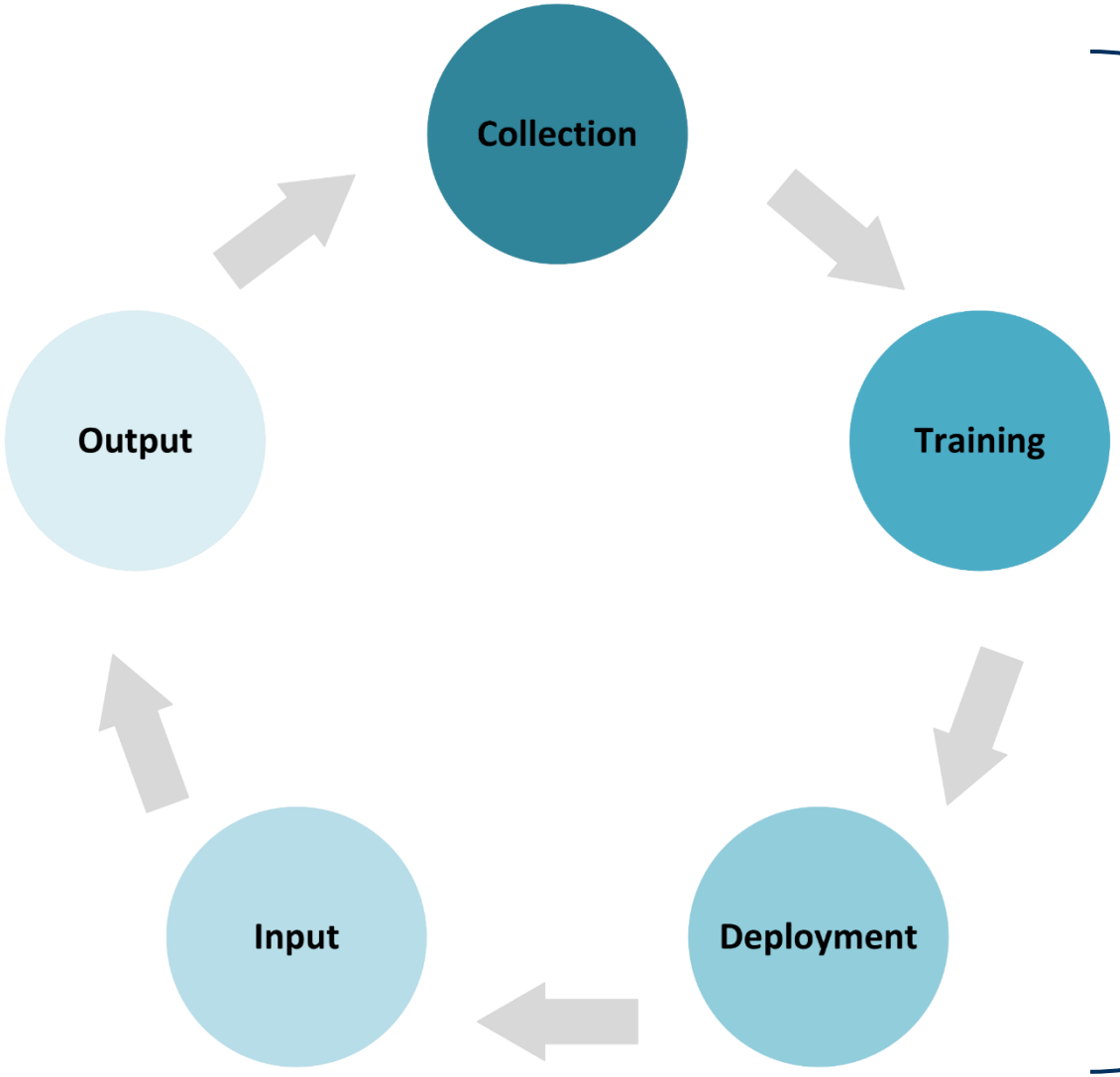


AI & Data Protection

Five Phases of Data Processing



Any information relating to an **identified or identifiable** individual is protected under the **GDPR**

⬇️

Legal basis for every data processing step necessary

Legal Basis for Public Entities

Public Entities:

✅ **Consent** (Art. 6 para. 1 lit. a GDPR)

⚠️ Informed, freely given, and explicit. Challenging for large-scale training data collection; revocability poses risks to AI functionality

✅ Performance of a **Contract** (Art. 6 para. 1 lit. b GDPR)

⚠️ Limited to processing necessary for contract performance; cannot cover third-party data

✅ Compliance with a **legal obligation** (Art. 6 para. 1 lit. c GDPR)

⚠️ Narrow scope; no discretion for the controller

✅ Protection of **vital interests** (Art. 6 para. 1 lit. d GDPR)

⚠️ Only for emergency/life-threatening situations; not for training

✅ Performance of a task carried out in the **public interest** or in the exercise of **official authority** (Art. 6 para. 1 lit. e GDPR)

⚠️ For public bodies, requires additional national legal basis

Legal Basis for Public Entities

- ❌ **Legitimate interests**, Art. 6 para. 1 lit. f DSGVO

  ➡️ Art. 6 para. 1 sub para. 2 DSGVO: „**Point (f)** of the first subparagraph **shall not apply to** processing carried out by **public authorities** in the performance of their tasks"

- **Employment** (§ 26 Federal Data Protection Act (BDSG)):
  2023 ECJ Case C-34/21:
  ❌ Sec. 26 BDSG **violates European law**
  ➡️ Performance of a **Contract** (Art. 6 para. 1 lit. b GDPR)

- **Scientific or historical research** and for statistical purposes (§ 27 Federal Data Protection Act (BDSG))

Legal Basis for Public Entities

- Public bodies in Sachsen-Anhalt:
  - General clause **(§ 4 DSAG LSA**)

    ⚠️ can only support **low-intensity processing**

    = minimal impact on the rights and freedoms of the data subjects

Gesetz zur Ausfüllung der Verordnung (EU) 2016/679 und zur Anpassung des allgemeinen Datenschutzrechts in Sachsen-Anhalt
(Datenschutz-Grundverordnungs-Ausfüllungsgesetz Sachsen-Anhalt - DSAG LSA)
Vom 18. Februar 2020*1)

Landesdatenschutzgesetz
(LDSG)
Vom 12. Juni 2018 *

§ 3a
Nutzung von KI-Systemen

Die Nutzung von KI-Systemen zur Verarbeitung personenbezogener Daten ist unbeschadet sonstiger Bestimmungen zulässig, wenn die Voraussetzungen für die Verarbeitung der personenbezogenen Daten als solche gegeben sind.

§ 4
Zulässigkeit der Verarbeitung personenbezogener Daten

Die Verarbeitung personenbezogener Daten ist zulässig, soweit sie erforderlich ist zur Erfüllung

1. einer rechtlichen Verpflichtung, der der Verantwortliche unterliegt, oder
2. einer in der Zuständigkeit des Verantwortlichen liegenden Aufgabe, deren Wahrnehmung im öffentlichen Interesse liegt oder die in Ausübung öffentlicher Gewalt, die dem Verantwortlichen übertragen wurde, erfolgt.

Im Übrigen bestimmt sich die Zulässigkeit der Datenverarbeitung nach Artikel 6 Abs. 1 der Verordnung (EU) 2016/679.

Profiling

Art. 22 GDPR:

"1. The data subject shall have the right **not to be subject** to a **decision based solely on automated processing**, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her.

[...]"

⬇️

ECJ Judgments of December 7, 2023, Case No. C-634/21 (Schufa Scoring)

❌ fully automated decisions
= decisions made without human involvement

✅ decision-support systems
= assist humans in making a decision; humans retain the final decision-making authority and can review and, if necessary, correct the system's

AI regulatory sandbox

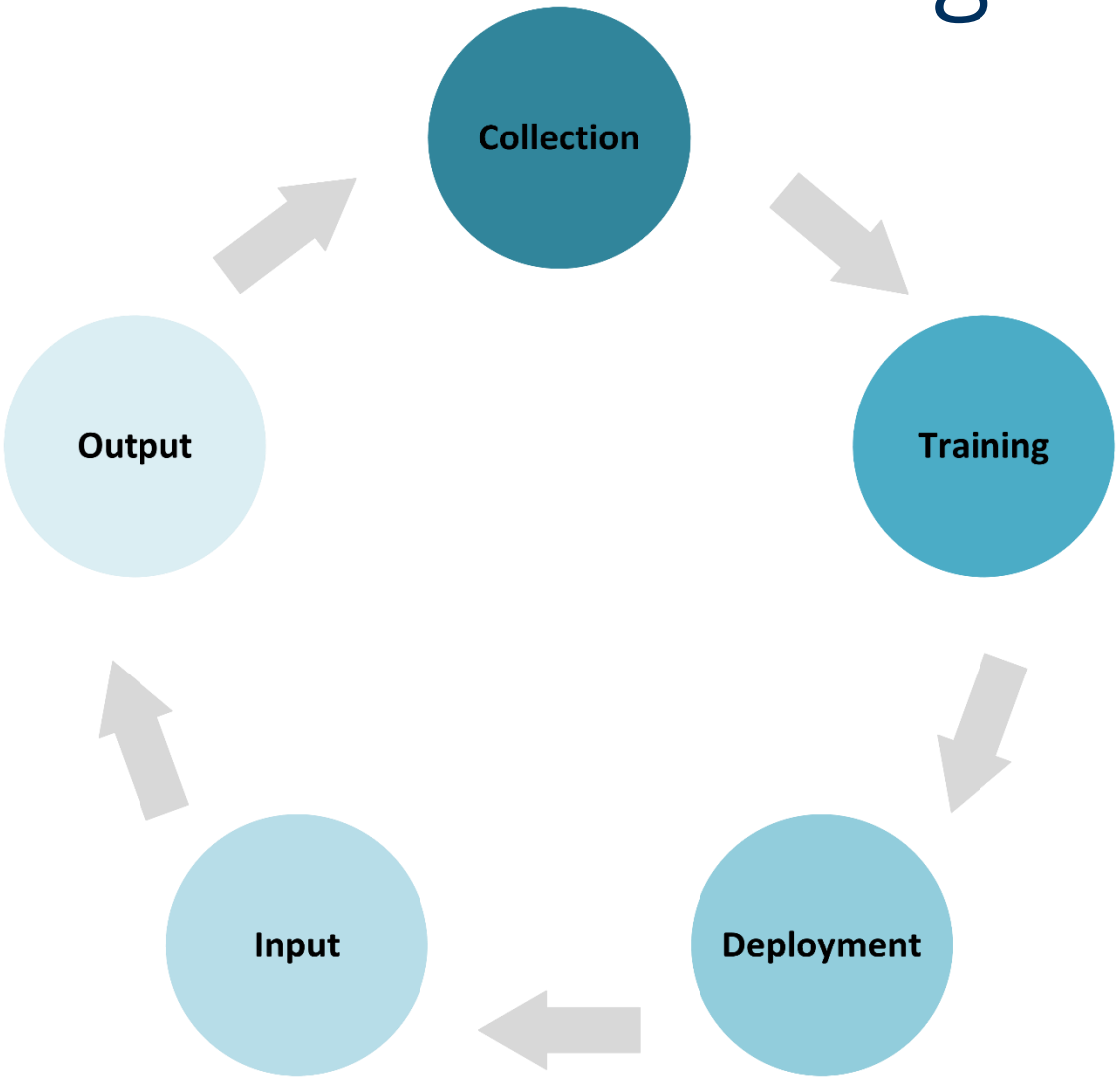
Art. 59 AI Act:

"1. In the **AI regulatory sandbox**, personal data **lawfully collected for other purposes** may be **processed** solely for the purpose of **developing, training and testing certain AI systems in the sandbox** [...]"

Regulatory Sandboxes
= controlled legal environments where people (usually employees, researchers, or entrepreneurs) can **experiment, take risks, and try out new ideas** can test new AI systems with real data but under relaxed regulations

AI & Copyright

Five Phases of Data Processing



(German) Copyright

- **Copyright** protects **original intellectual creations** (works)
  - Only **individual, creative, human made** achievement are protected
  - **Thoughts, ideas, concepts, facts** are **not protected**, only their presentation
- **author** has **exclusive right** to
  - **use the work** and
  - allow others to **use the work** and determine under what **conditions**
- ⚠️ under certain conditions **exceptions** of the German Copyright Act **permit the use** of others' works **without their consent**

(German) Copyright - AI Training

https://tu-dresden.de/robots.txt

```
User-agent: *
Disallow: /@@search$
Disallow: /@@search?
Disallow: /*/@@search$
Disallow: /*/@@search?
Disallow: /search$
Disallow: /search?
Disallow: /*/search$
Disallow: /*/search?
Disallow: /RSS$
Disallow: /RSS?
Disallow: /*/RSS$
Disallow: /*/RSS?
Disallow: /rss.xml$
Disallow: /rss.xml?
Disallow: /*/rss.xml$
Disallow: /*/rss.xml?
Disallow: /atom.xml$
Disallow: /atom.xml?
Disallow: /*/atom.xml$
Disallow: /*/atom.xml?
Disallow: /ics_view$
Disallow: /ics_view?
Disallow: /*/ics_view$
Disallow: /*/ics_view?
Disallow: /Shibboleth.sso/
```

AI Training

Sec. 44b
Text and data mining
Act on Copyright and Related Rights
(Urheberrechtsgesetz – UrhG)

- ✅ **Reproduce lawfully accessible** works
- ✅ **Copies** are **to be deleted** when no longer needed
- ✅ No **machine-readable usage restriction**

Sec. 60d
Text and data mining for scientific research purposes
Act on Copyright and Related Rights
(Urheberrechtsgesetz – UrhG)

- ✅ **Reproduce lawfully accessible** works
- ✅ **Scientific res**earch purposes
- ✅ **Research organisations**

First landmark rulings in Germany (as of 03.2026)

1. Munich Regional Court

LG München I,11.11.2025 - 42 O 14139/24

2. Hamburg Higher Regional Court

OLG Hamburg, 10.12.2025 - 5 U 104/24

Parties: GEMA (Plaintiff) vs. OpenAI (Defendant)
Outcome: GEMA wins – OpenAI must cease and desist, provide information, and pay damages
Details:

- OpenAI trained ChatGPT using nine well-known German song lyrics (including “Atemlos,” “Männer,” and “Über den Wolken”).These song lyrics can be retrieved almost verbatim using simple prompts. OpenAI denied having stored the lyrics and argued that the model “only learns, it does not store.”

Legal reasoning of the court:

- Retention of information affirmed: The court relies on IT research and finds that training data (song lyrics) are permanently stored in the model parameters. The reproducibility of the lyrics rules out coincidence.
- **§ 44b UrhG** does not apply: The exception permits **only preparatory analytical steps**, but **not permanent reproductions within the model**. Analogous application was rejected.

First landmark rulings in Germany (as of 03.2026)

1. Munich Regional Court

LG München I,11.11.2025 - 42 O 14139/24

2. Hamburg Higher Regional Court

OLG Hamburg, 10.12.2025 - 5 U 104/24

Parties: Photographer (Plaintiff) vs. LAION e.V. (Defendant)
Outcome: The defendant's use of the photograph is permitted under copyright exceptions (Sections 44b, 60d of the German Copyright Act).
Content:
The defendant makes a dataset of image-text pairs publicly available free of charge for the purpose of training generative AI models.The dataset contains hyperlinks to publicly accessible images on the internet as well as additional information about the images, including image descriptions (alt text) and metadata.Use of a photograph by the plaintiff in the freely accessible AI training dataset LAION 5B
Legal reasoning of the court:

- **§ 44b UrhG does apply**: The automated **analysis** and **reproduction** of the photo **for the creation of an AI training dataset** falls under the TDM exception. At the time of the download, the **usage restriction** on the image provider's website was not **machine-readable**
- **§ 60d UrhG**: The creation of the dataset served **scientific research purposes**. Even if the dataset may later be used by commercial entities, the research purpose remains decisive as long as there is no controlling influence by private companies.

(German) Copyright – Input & Output

Input

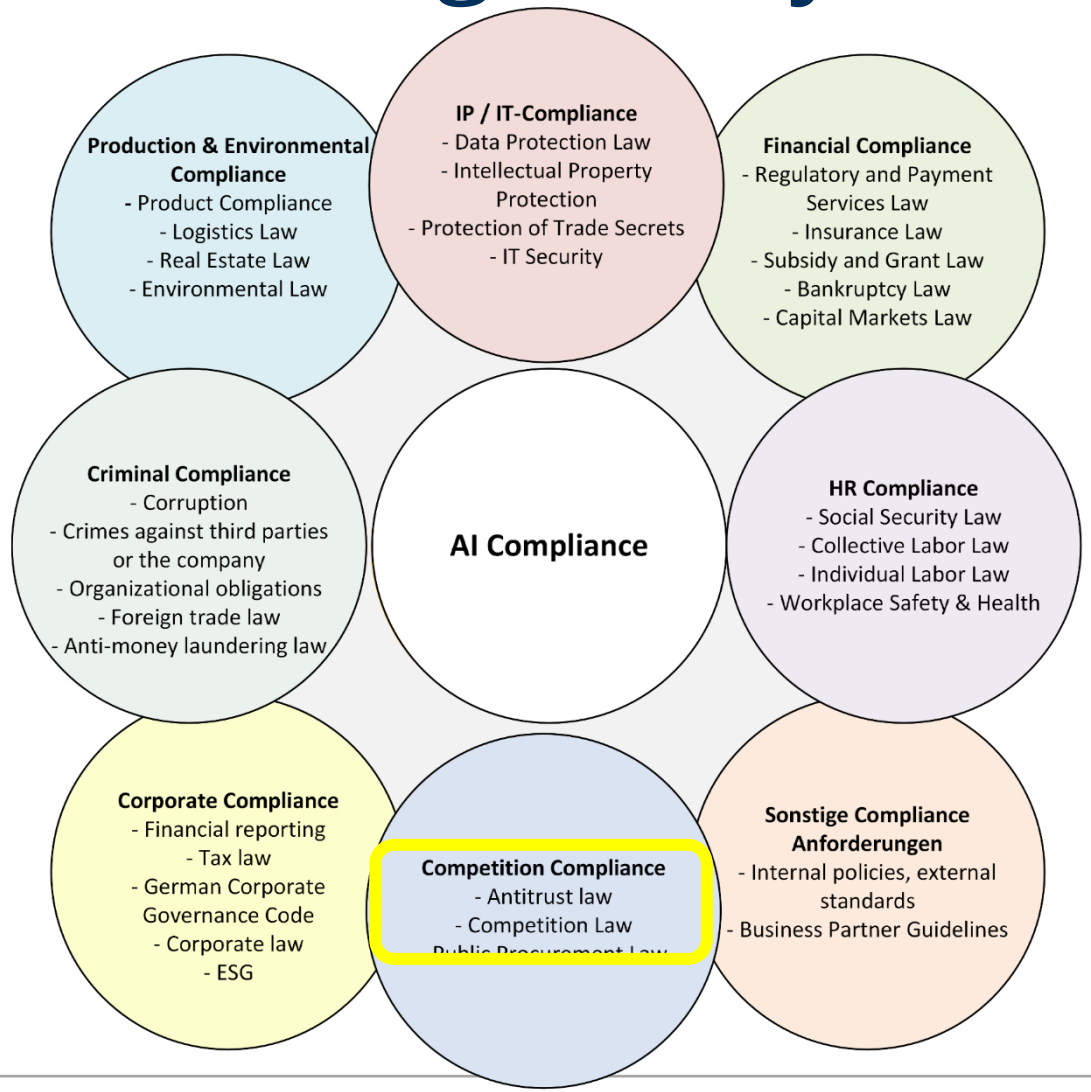
- Do not upload copyrighted works created by others!
  - ⚠️ **Reproduction** (Sec. 16 German Copyright Act)
- ➡️ Use only own works or works with a free license
- ➡️ Formulate prompts such as: “Generate … in the style of …”

Output

- AI-generated output is **not protected** by copyright
  - ❗ Copyfraud (Sec. 97 German Copyright Act)
- Disclosure requirement:
  - University guidelines
  - Art. 50 (4)
- Review output

AI & Competition law

Chapter 3: Other Regulatory Framework



AI & Competition law

PRESS RELEASE | Dec 9, 2025 | Brussels | 3 min read

Commission opens investigation into possible anticompetitive conduct by Google in the use of online content for AI purposes

„The European Commission has opened a formal antitrust investigation to assess whether **Google** has breached EU competition rules by **using the content of web publishers**, as well as **content uploaded on** the online video-sharing platform **YouTube, for artificial intelligence (‘AI') purposes.”**

https://ec.europa.eu/commission/presscorner/detail/de/ip_25_2964

Thank You :)

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